

Graduation Guide

MSc Geomatics

v2025–2026

Preface

The graduation phase is the last step in the MSc Geomatics programme. It is completed in the form of an individual research project.

The topic of the scientific project can be related to any of the subjects offered by the MSc Geomatics programme.

The project can be performed in cooperation with a company/institution, which provides a use case or a problem statement (but the work is always supervised by the scientific staff of the university).

More information, news, updates, downloaded templates, and a calendar can be found on the official website: <https://geomatics.bk.tudelft.nl/geo2021/>.

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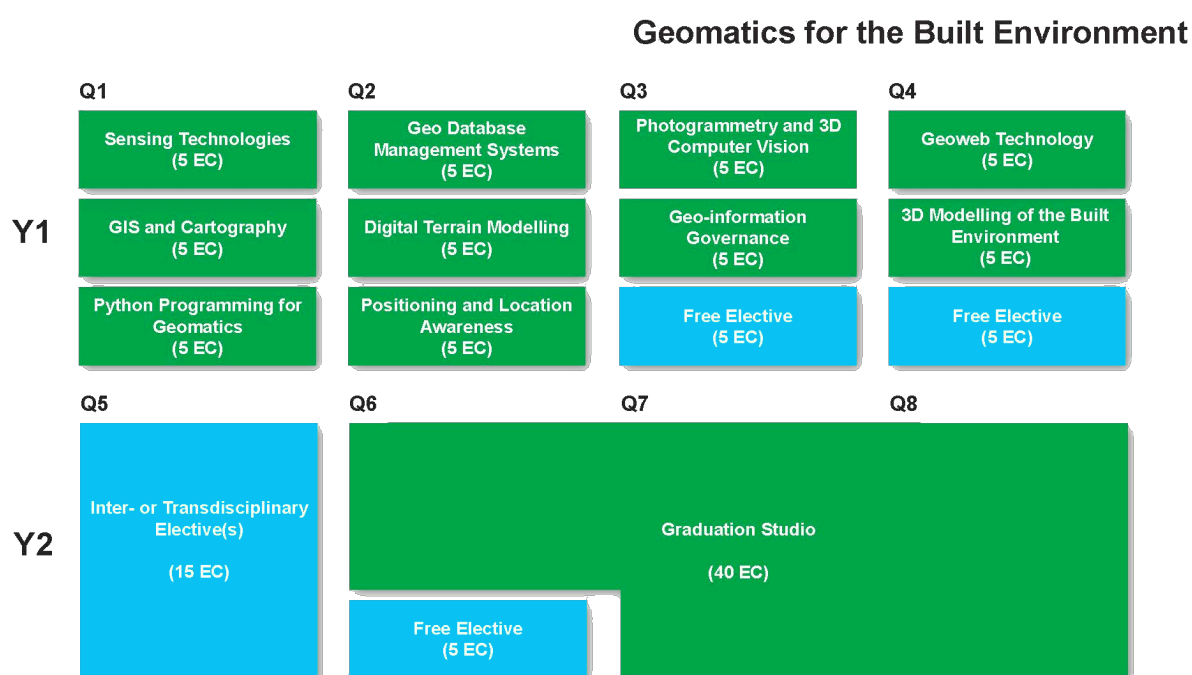
1. Graduation structure

1.1. Duration

The *Geomatics Graduation Studio* (the graduation phase where the MSc thesis is written; GEO2021) is 40ECTS and lasts in general three quarters.

It normally takes place during the last three quarters of the second Master's year (quarters Q6+Q7+Q8), but it can be started twice during the academic calendar.

1.2. Structure



1.3. Admission and assessments

To enter the graduation phase (at A1), the student must have obtained 55 ECTS from the first year of the programme, and have followed the Q5 electives (15 ECTS); see image structure above.

The graduation phase is divided into the following assessments:

1.4. Calendar

All assessments (except A2) should be held in the weeks indicated below. Please be aware that there are deadlines for the registration for each assessment. For the academic year 2025-2026, this implies the following weeks and deadlines, as registered in the BK academic graduation calendar:

Assessment	Name	Details
A0	Registration	Selection of the first supervisor and of a thesis topic, either by picking from a topic from the list or by agreeing on a custom topic.
A1	Kick-off	Formal ‘go/no go’ assessment of the graduation plan, 15 min presentation and discussion.
A2	Midterm	A structured meeting with both supervisors to discuss progress and updated plans. The form is decided by the supervisors.
A3	Green light	Formal final assessment of the complete graduation report (MSc thesis), presentation of ~20min, and discussion.
A4	Finalisation	Presentation for a lay audience.

Academic Calendar 2025 / 2026

Graduation

Autumn semester

Calendar Week	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	1	2	3	4			
Teaching week	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10	2.1	2.2	2.3	2.4	2.5	2.6 Christmas period					2.7	2.8	2.9	2.10
	Sept.				Oct.				Nov.				Dec.				Jan.							
Mon	1	8	15	22	29	6	13	20	27	3	10	17	24	1	8	15	22	29	5	12	19	26		
Tues	2	9	16	23	30	7	14	21	28	4	11	18	25	2	9	16	23	30	6	13	20	27		
Wed	3	10	17	24	1	8	15	22	29	5	12	19	26	3	10	17	24	31	7	14	21	28		
Thurs	4	11	18	25	2	9	16	23	30	6	13	20	27	4	11	18	25	1	8	15	22	29		
Fri	5	12	19	26	3	10	17	24	31	7	14	21	28	5	12	19	26	2	9	16	23	30		

Spring semester

Calendar Week	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
Teaching week	Spring break	3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	3.10	4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	4.10
Feb.	2	9	16	23	2	9	16	23	30	6	13	20	27	4	11	18	25	1	8	15	22
Tues	3	10	17	24	3	10	17	24	31	7	14	21	28	5	12	19	26	2	9	16	23
Wed	4	11	18	25	4	11	18	25	1	8	15	22	29	6	13	20	27	3	10	17	24
Thurs	5	12	19	26	5	12	19	26	2	9	16	23	30	7	14	21	28	4	11	18	25
Fri	6	13	20	27	6	13	20	27	3	10	17	24	31	8	15	22	29	5	12	19	26

Summer period

Calendar Week	27	28	29	30	31	32	33	34	35
Summer period	5.1	5.2	5.3	5.4	5.5	5.6	5.7	5.8	5.9
July	29	6	13	20	27	3	10	17	24
Tues	30	7	14	21	28	4	11	18	25
Wed	1	8	15	22	29	5	12	19	26
Thurs	2	9	16	23	30	6	13	20	27
Fri	3	10	17	24	31	7	14	21	28

Day colours legend



Week colours legend



Final registration date for A1. Assessment for next A1 period.

Final registration date for A3 and A4 assessments in Q4.

No education

A1 Kick-off assessments

Advised weeks for A2 assessments - halfway A1 and A3

Both A1 Kick off assessment + A4 Final assessment

A3 Green light assessment

A4 Final assessment

A4 - Public final presentations take place two weeks after the A3

Public Holidays

Christmas period	Dec 22 - Jan 2
Spring Break	Feb 2 - Feb 6
Good Friday	April 3
Easter	April 5 & 6
Kings Day	April 27
Liberation Day	May 5
Ascension Day	May 14
Whit Monday	May 25

1.5. Retakes

All formal As are offered in every quarter.

At the A1 (Kick-off), the supervisors with the delegate determine whether the student's plan and presentation meet the requirements, resulting in a 'go' to the A2 (Midterm) phase. If a 'no-go' is given, the A1 can be done again in the next quarter.

The same applies to the A3 (Green light), where the supervisors, co-reader, and the delegate give a 'go' to proceed to the public A4 (Finalisation) two weeks later (the grades are also determined but not yet communicated); if the requirements are not met, the consequent 'no go' results in a retake of the A3 in the next quarter.

1.6. Registration

For planning purposes, the faculty is using the *SuperSaaS* tool, which can be accessed by the student's supervisors. It is the responsibility of the responsible supervisor to register the student's A1, A3 and A4 applications before the deadlines in the BK academic graduation calendar. This makes it possible to have a room for the assessment, as well as for your supervisors, co-reader and delegate to record a timely reservation in their agenda. If dates need to be changed, or if dates in SuperSaaS are not available for the quarter, this should be resolved within the supervisory team and communicated to Education and Student Affairs (E&SA) (graduation-bk@tudelft.nl).

For registering the graduation process, the faculty is using the *MyCase* registration tool (see Appendix I). It is the student's task to start their own 'graduation case'. MyCase is used for the registration of assessment dates, submission of the deliverables before an assessment, registration of the assessments, and feedback. Moreover, the names of the supervisors, co-reader and delegate, and the progress and assessment results are registered in this application. After each assessment, the responsible supervisor fills in the assessment (see Chapter 4 and Appendix IV).

1.7. Ethics

In projects involving humans, certain types of data processing increase the risks to the human participants: planning such projects requires additional evaluations and advice from university staff before ethical approval can be received and the project can begin. In the case of a graduation project, obtaining additional advice or permits may delay the project with an extra semester. Therefore, all students have to check their risk, by completing the Ethics Checklist (Appendix VI) before the A1. Only if the graduation project involves working with data from human participants, should the rest of the checklist be completed and contact be made with the Human Research Ethics Committee (HREC).

2. Supervisory team

The student's supervisory team is composed of two supervisors and a co-reader. The supervisors guide the student during the graduation process and are responsible for assessing the student's work and progress at the A1 (Kick-off) and A2 (Midterm). Together with the co-reader, who serves as an independent expert, they also assess and determine the grades at the A3 (Green light), which are only disclosed at the A4 (Finalisation), as seen in Chapter 4.

2.1. Supervisors

The responsible supervisor (also called first or main supervisor) is an expert in the graduation project field and must be a scientific staff member (including PhD students and postdocs) from the groups involved in MSc Geomatics education: 3D geoinformation, GIS technology (GDMC), or the Knowledge center open data. Students choose this supervisor during the A0 (Registration) phase, either by directly contacting them to agree on a custom topic or by selecting a predefined topic linked to the supervisor.

The second supervisor is a scientific staff member from TU Delft whose expertise complements that of the responsible supervisor. She/he may be predefined based on the selected topic or chosen by the responsible supervisor with input from the student. If the responsible supervisor doesn't hold a PhD degree, the second supervisor must have one. Ideally, the second supervisor is determined at the A0 (Registration) phase and must be confirmed before the A1 (Kick-off).

2.2. Co-reader

The co-reader is a scientific staff member of TU Delft or another university who is an expert in the field of the graduation project. Their role is to assess the quality of the student's work in an unbiased way at the A3 (Green light). Preferably, they are not part of the same group(s) as the supervisors. The co-reader is chosen by the supervisors in collaboration with the student. The co-reader is preferably defined at the A0 (Preparation) and must be defined before the A3 (Green light).

2.3. Delegate of the Board of Examiners

Since the Board of Examiners is responsible for guarding the quality of all examinations, a delegate is appointed by the administration of the Faculty before the A1: a (former) teacher who is familiar with the graduation process and who is not from the same section as your supervisors; this to assure independence. The delegate functions independently from the supervisors and student, and chairs the A1 (Kick-off) and A3 (Green light). In that capacity, the delegate is there to protect the student against process errors and possibly unfair treatment by the supervisors, while on the other hand ensuring that the student is assessed according to the appropriate criteria by the supervisors (Chapter 4).

3. Scientific approach

3.1. Research project

The graduation project is an individual research project that is conducted by the student with guidance and regular feedback from the supervisors.

The project starts with a preliminary phase where the student lays down the foundations for their research, which includes conducting a study of the most important relevant literature, refining their topic into a specific problem statement, establishing of clear research questions or objectives, and selecting an appropriate methodology. This culminates into the creation of a graduation plan (see Appendix III), which is presented, discussed assessed at the A1 (Kick-off) meeting.

Upon approval of the graduation plan, the student conducts their research according to the methodology and time planning in it. The specific contents of this phase of the graduation will vary depending on the chosen methodology, but might include among others: an extended survey of scientific literature and other sources, data collection and analysis, developing new methods, tools or datasets, and software development and testing.

The main deliverable of the graduation project is a written graduation report (see Appendix III), which documents the research that was conducted. It includes a reflection that is integrated into the report, mostly in the introduction and conclusions. The contents of the report will differ depending on the nature of the research, but they should be detailed enough to allow for reproducibility of the work presented in it. Similarly, in the interest of reproducibility, if the research involved the creation of source code or data, students are highly encouraged to publish these in a persistent public repository which is linked in the report.

In addition to the graduation report, there are two other presentations as deliverables: (1) a technical presentation, which is given, discussed and assessed at the A3 (Green light) meeting, and (2) a presentation for a lay audience, which is given, discussed and assessed at the A4 (Finalisation) meeting.

3.2. Graduation plan (A1) structure

The graduation plan for A1 is a project proposal that must contain the following elements:

1. A **cover page** with
 1. title
 2. full name
 3. student number
 4. names of supervisors #1 + #2
2. An **introduction** in which the relevance of the project and its place in the context of geomatics is described, along with a clearly-defined problem statement;
3. A **related work** section in which the relevant literature is presented and linked to the project;
4. **Research questions** that are clearly defined, along with the scope (ie what the student will and will not be doing);

5. Overview of the proposed **methodology** to be used;
6. Overview of **preliminary results** (if application, discuss with your supervisors);
7. **Time planning**—having a Gantt chart is probably a better idea than just a list;
8. Since specific **data and tools** have to be used, it's good to present these concretely, so that the supervisors know that you have a grasp of all aspects of the project;
9. The **references**.

Supervisors expect a document of 10–15 pages.

A LaTeX template is available in Appendix III.

3.3. Graduation report (A3) structure

The exact structure of the thesis, or graduation report, will depend on the type of work that is involved. However, most good Geomatics theses contain something similar to the following elements:

1. Title page, possibly split into cover page, half-title, copyright page (with licence and names of supervisors and co-reader), etc;
2. Abstract;
3. Acknowledgements;
4. Table of contents, possibly accompanied with lists of figures, tables, algorithms, acronyms, etc;
5. Introduction, including motivation, research questions and a summary of all chapters;
6. Related work, possibly preceded by a background chapter;
7. Methodology, possibly followed by implementation and/or data chapters;
8. Results, including a critical evaluation and discussion of them;
9. Conclusions and future work.

The implications and reflection of the work that was conducted should be integrated throughout the thesis. In particular, the implications should be present in the introduction and conclusions, and the reflection should be present in the conclusions.

A LaTeX template is available in Appendix III.

4. Feedback and assessment

Throughout the graduation phase, the student should receive regular feedback from the supervisory team. More extensive feedback is provided during the A1, A2 and A3 assessments.

One week before the A1 and A3 assessments, the student submits the main deliverables (graduation plan for A1 and graduation report for A3). These are presented at the A1 and A3 assessment sessions, where the independent delegate chairs the session. Following the presentation, the student responds to the supervisory team's questions. Then, the supervisory team and delegate retreat for deliberation, after which the results are announced and feedback is provided.

During the A2 assessment, the supervisors and student conduct a more structured meeting to discuss the progress and (updated) plans. The form of this session depends on the supervisors, but it might include a presentation, demo or extended discussion session.

At the A1 and A3, the result is formally expressed in a 'go' or a 'no-go'. In case of a 'no-go', a retake of the assessment can take place in the next education period (see Chapter 1 and Appendix II). At the A3, the 'go' or 'no-go' determines both whether the student continues to the A4 and the preliminary grades, which will not yet be disclosed (see below).

4.1. Components of assessments

For the A1 and A2, it is being assessed whether the work is quantitatively and qualitatively on schedule with respect to the A3.

4.2. Deliberation and feedback

After the A1 and A3 presentations and subsequent discussion, the supervisory team will retreat for deliberation. At the A1 and A3, the independent delegate chairs the deliberation (see Chapter 2 and Appendix II), without influencing the result.

During the deliberation, the supervisory team assesses the deliverables, including the presentation and the answers to the supervisory team's questions. This assessment is formally expressed in a 'go' or a 'no go', representing a 'sufficient to good' or 'insufficient' level respectively.

At end of the deliberation for the A1 and A3, the responsible supervisor should communicate the results of the deliberation ('go' or 'no go') and provide feedback to the student, including how to improve. At the A2, there might not be a deliberation, but the supervisors should still provide feedback to the student. Afterwards, the responsible supervisor should register this result and feedback in MyCase.

4.3. Grades

The grades that are determined at the A3 are provisional will not yet be announced during the A3. Two weeks later, at the public A4, the final grade list will be presented—but not publicly disclosed—to the student together with the diploma. In exceptional cases, based on the quality of the lay presentation, the preliminary final grade can be increased or lowered by half a point by the supervisory team (see Chapter 5).

5. Public presentation (A4)

5.1. Lay audience

The student's A4 takes place two weeks after the A3 (Green light). This is a public session for a lay audience where family, friends, and fellow students can be invited.

Since the student is presenting to a different kind of audience, they need to adapt their A3 presentation accordingly. However, the thesis remains unchanged, as the preliminary grades were already determined at A3.

5.2. Quality of the lay presentation

The ability to communicate with non-specialists is explicitly listed in the final attainment levels (see Appendix VII). Since this competence isn't evaluated elsewhere during graduation, the A4 provides an ideal opportunity to demonstrate this skill.

Following the A4, the assessment committee will evaluate whether the lay presentation demonstrates "exceptional quality" or falls "below the expected quality" using the specific A4 rubric (see Appendix IV). Based on this assessment, the provisional final grade may be adjusted up or down by half a point. Note that this adjustment doesn't count toward the cum laude distinction (see below), and if the provisional final grade is 6.0, it cannot be lowered.

5.3. Cum laude

Based on the results gained for the courses in the Master's Programme and the final thesis, the student can be awarded a cum laude distinction. A student will receive this if all of the following requirements are met:

1. The weighted average of the grades of the courses, not including the graduation project, must be at least 8.0 (exemptions ("VR" in OSIRIS) will not be taken into consideration; no more than 20 ECTS may be completed with an exemption);
2. No courses may be completed after a repair resulting in a 'pass';
3. The provisional final grade of the graduation project at the A3 must be at least 8.5;
4. The study duration cannot exceed the nominal study duration plus one semester.

For students who do not meet all criteria above, especially the fourth, for instance because they were enrolled as a student in a specific semester but did not participate in any course or examination in that semester, an exemption may apply. This should be arranged before the A3 application date (by sending a request by email to the Board of Examiners: BoardofExaminers-BK@tudelft.nl).

I. MyCase registration

As explained in Chapter 1, *MyCase* is used as a system to track the graduation progress of each student. All information about the graduation (general information and planning, stakeholders, documentation and assessment) is collected in this system. The table below serves as a chronological overview of all tasks and responsibilities for students who start their thesis in Q1.

Phase	When	What	Who
A0 Registration	1.3	Submit thesis topic form (on GEO2021 website)	Student
	1.8–2.2	Register new case in My-Case, including supervisors and planning in weeks	Student
		Register A1 date in Super-SaaS*	Responsible supervisor
		Check entry requirements	Faculty administration
	2.5	Approve supervision team	GEO2021 coordinator
	2.6	Register delegate by BoE	MyCase faculty administration
A1 Kick-off	2.8–2.9	Upload graduation plan, human participation, external party and confidentiality agreement, and submit ‘Ready for Kick-off’ task in My-Case	Student
	2.9–2.10	Kick-off assessment	Student, supervisors and delegate
		Submit result and feedback in MyCase, register dates of A3 and A4 in SuperSaaS*	Responsible supervisor
A2 Midterm	3.8–3.9	Upload midterm materials, and submit ‘Ready for Midterm’ task in MyCase	Student
	3.9–4.0	Midterm assessment	Student and supervisors
		Submit result and feedback in MyCase	Responsible supervisor
A3 Green light	4.6–4.7	Check study progress, upload graduation report, submit diploma application (with final title!) and submit ‘Ready for green light’ task in My-Case, and upload graduation	Student

Phase	When	What	Who
		report in plagiarism scan in Brightspace	
		Check graduation requirements	Student programme administration
	4.7–4.8	Check for plagiarism	Responsible supervisor
		Green light assessment	Student, supervisors, co-reader and delegate
		Submit result and feedback in MyCase	Responsible supervisor
A4 Finalisation	4.9–4.10	Upload lay presentation and submit 'Ready for finalisation' task in MyCase, upload graduation report and lay presentation in repository	Student
		Upload final grade	First supervisor

Information

If a student does not start the graduation at Q2 or Q4, or if they are delayed, SuperSaaS (the scheduling tool used by the administration) might not be available. In such cases, A1/A3/A4 should be planned by sending an email to graduation-bk@tudelft.nl.

II. Assessments (A/0/1/2/3/4)

II.i. A0 (Registration)

Period	Every quarter this can be done, ask the coordinator of GEO2021
Actions student	- Find and agree on a topic with a MSc Geomatics staff - Write a short summary - Upload the required information to the GEO2021 website

II.ii. A1 (Kick-off)

Admission requirements	The student must have obtained 55 ECTS from the first year of the programme, and have followed the Q5 electives (15 ECTS).
Period	In fixed weeks, once per quarter
Preparation	
Planning	Done by the responsible supervisor in SuperSaaS. If not available, by email to graduation-bk@tudelft.nl
Location	Room booked by scheduling department
Deliverables	Project proposal
Assessment method	Rubric
Actions student	- Upload graduation plan to MyCase (1-week before) - Prepare presentation
Actions supervisors	Read Graduation Report before start A1
Actions delegate	Read Graduation Report before start A1
At the session	
Schedule	15 min: preparation by student (make sure projector working), 15 min: presentation by student, 15 min: questions by supervisors, 15 min: deliberation and feedback
Assessors	First and second supervisors
Quorum	Both supervisors and delegate, in person
Actions supervisors	Fill in the feedback form, upload it in MyCase and give feedback
Actions delegate	Chair the assessment and the deliberation
Upon completion	
Assessment result	<i>Go</i>: the student proceeds towards the A2 <i>No-go</i>: the student agrees a new A1 date for the next A1 period with the supervisors and delegate; the first supervisor makes a reservation in the SuperSaaS registration programme. If not available, by email to graduation-bk@tudelft.nl <i>Withdrawal</i>: same as for no go.

Assessment registration	The feedback and assessment ('Go' or 'No-go') is registered in MyCase by the first supervisor
Planning A3 and A4	In case of a positive result, the responsible supervisor agrees the A3 and A4 date with the other supervisor, delegate and student, and registers both dates in the SuperSaaS registration program. If not available, by email to graduation-bk@tudelft.nl .

II.iii. A2 (Midterm)

Admission requirements	The student successfully passed A1
Period	Anything between A1 and A3, it is up to the student and the first supervisor to agree on a date
Preparation	
Planning	Done by main supervisor
Location	Room booked by the first supervisor
Deliverables	A document must be uploaded before the meeting to MyCase. The form/content of this document is decided by the supervisors.
Assessment method	(none)
Actions student	Upload the required document to MyCase (at last 24h before A2)
Actions supervisors	First supervisor books the room and invite the second supervisor
At the session	
Schedule	Meeting has free form, it is decided by the supervisors
Assessors	(this is not a formal assessment)
Quorum	Both supervisors
Actions student	(none)
Actions supervisors	(none)
Upon completion	
Assessment result	(an advice is given to the student, based on their progress)
Assessment registration	The feedback/advice is registered in MyCase by the first supervisor

II.iv. A3 (Green light)

Admission requirements	The student must have completed all Master's courses with exception of the graduation project, must have successfully completed assessment A1 of the graduation, and must have held assessment A2 of the graduation.
Period	In fixed weeks, once per quarter.

Preparation	
Planning	Day and timeslot reservation made by first supervisor after the A1
Location	Room booked by scheduling department
Deliverables	A full thesis and a presentation
Assessment method	Rubric
Actions student	• Upload final draft thesis to MyCase (1-week before), • Perform the plagiarism scan in Brightspace • Prepare presentation
Actions supervisors	• Read Graduation Report before A3 start, • Check if and how student implemented advice/feedback from A1 and A2.
Actions delegate	Skim (or read) the Graduation Report before A3 start
At the session	
Schedule	• 15 min: preparation by student (make sure projector working) • 20 min: presentation by student • 20 min: questions by supervisors • 15 min: deliberation and feedback
Assessors	First and second supervisors, and the co-reader
Quorum	Both supervisors, co-reader, and delegate, in person
Actions student	Give presentation and answer questions
Actions supervisors	Fill in the feedback form, upload it in MyCase and give feedback
Actions delegate	Chair the assessment and the deliberation
Upon completion	
Assessment result	• <i>Green-light</i>: the student proceeds towards the A4 • <i>Red-light</i>: the student agrees a new A3 date for the next A3 period with the supervisors and delegate; the first supervisor makes a reservation in the SuperSaaS registration program. If not available, by email to graduation-bk@tudelft.nl. • <i>Withdrawal</i>: same as for no go.

Assessment registration	The feedback and assessment ('go' or 'no go') is registered in MyCase by the first supervisor
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II.v. A4 (Finalisation)

Admission requirements	The student must have successfully completed assessment A3 of the graduation.
Period	In fixed weeks, once per quarter. Must be done 2 weeks after the A3, a student cannot postpone A4 after a successful A3.
Preparation	
Planning	Day and timeslot reservation made by first supervisor after the A1
Location	Room booked by scheduling department
Deliverables	Presentation
Assessment method	Rubric
Actions student	• Upload final thesis to the TU Delft repository (1 day before A4), • Prepare presentation for lay audience
Actions supervisors	(none)
Actions delegate	(none)
At the session	
Schedule	• 15 min: preparation by student (make sure projector is working) • 15 min: presentation by student • 15 min: questions by supervisors • 15 min: deliberation and feedback
Assessors	First and second supervisors
Quorum	Both supervisors, in person
Actions student	• Give presentation and answer questions • Sign the diploma on both sides
Actions supervisors	• Fill in the A4 rubrics and determine whether the provisional final grade should be increased or lowered by half a point • Fill in the final grade list • Hand over the diploma and the grade list (the latter in a closed envelope), and give the laudation
Upon completion	
Assessment registration	The final grade is registered in MyCase by the first supervisor
Actions student	• Unsubscribe as TU Delft student via Studielink enrolment, so your enrolment will be terminated as from the 1st of the next month (if you do not terminate your enrolment in time you are required to pay tuition fees for another month;

	retroactive termination of your enrolment is not possible; under certain circumstances the tuition fee can be partly refunded, see the website Termination of Enrolment) • Fill in exit evaluation (you will receive an email).
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III. Deliverables

III.i. A1 (project proposal) template

```

\documentclass[a4paper,11pt]{scrartcl}

\usepackage{graphicx}
\usepackage[utf8]{inputenc} %-- pour utiliser des accents en français
\usepackage{amsmath,amssymb,amsthm}
\usepackage[round]{natbib}
\usepackage{url}
\usepackage{xspace}
\usepackage[left=20mm,top=20mm]{geometry}
\usepackage{algorithmic}
\usepackage{subcaption}
\usepackage{mathpazo}
\usepackage{booktabs}
\usepackage{hyperref}

%-- shortcuts
\newcommand{\ie}{ie}
\newcommand{\eg}{eg}
\newcommand{\reffig}[1]{Figure~\ref{#1}}
\newcommand{\refsec}[1]{Section~\ref{#1}}

\setcapindent{1em} %-- for captions of Figures

\renewcommand{\algorithmicrequire}{\textbf{Input:}}
\renewcommand{\algorithmicensure}{\textbf{Output:}}

\title{My graduation plan (A1) template}

\author{
  Céline Dion\\
  student \#123456 \\
  \url{c.dion@tudelft.nl} !!!USE TUDELFT EMAIL, NOT PRIVATE EMAIL!!!\\
  \\
  1st supervisor: Jan Smit \\
  2nd supervisor: Gerard Joling \\
}

\date{13 January, 2024}

\begin{document}

\maketitle
\newpage

%%
%
\section{Introduction}
An introduction in which the relevance of the project and its place in the

```

context of geomatics is described, along with a clearly-defined problem statement.

%%%

%

`\section{Related work}`

A related work section in which the relevant literature is presented and linked to the project.

%%%

%

`\section{Research questions}`

The research questions are clearly defined, along with the scope (ie what you will not be doing).

To help you define a ``good" research question, read `\url{https://geomatics.bk.tudelft.nl/geo2021/templates/Research-Questions_WS-handout.pdf}`.

% `\citet{Smith03}` cites something as a noun, and it's also possible to put the references between parentheses at the end of a sentence~`\citep{Smith03}`

%%%

%

`\section{Methodology}`

Overview of the methodology to be used.

%%%

%

`\section{Preliminary results}`

If you have preliminary results, you can show them here.

This shows that you have actually started the work and you have not only been reading.

%%%

%

`\section{Time planning}`

Having a Gantt chart is probably a better idea than just a list.

%%%

%

`\section{Tools and datasets used}`

Since specific data and tools have to be used, it's good to present these concretely, so that the mentors know that you have a grasp of all aspects of the project.

`\bibliographystyle{abbrvnat}`

`\bibliography{myreferences}` %-- path of myreferences.bib

`\end{document}`

(notice that this text is available at <https://gist.github.com/hugoledoux/d16d5a4d397858ac745e38f9e8561657>)

III.ii. A3/A4 report (final thesis)

We offer two templates (one in LaTeX and one in Typst), those are available on the website of GEO2021: <https://geomatics.bk.tudelft.nl/geo2021/templates/#thesis-template>

III.iii. A3 presentation

The A3 (Green light) presentation is a formal assessment where the student presents the complete findings of their MSc thesis to their supervisory team. The content is a technical summary of the entire graduation report, covering the research questions, methodology, results, and conclusions.

The presentation lasts 20 minutes and is followed by a 20-minute question session from the assessment committee. The audience consists of the first and second supervisors, the independent co-reader, and a delegate from the Board of Examiners.

The goal of this presentation is to demonstrate that the thesis work is complete and of sufficient quality to proceed to the final public defense (A4).

III.iv. A4 presentation

The A4 “Finalisation” presentation is the public defense of the MSc thesis for a lay audience. This includes family, friends, and fellow students.

The content must be an adapted version of the A3 presentation, translated to be understandable for non-specialists. The presentation lasts 15 minutes and is followed by a 15-minute question session.

The goal is to demonstrate the ability to communicate complex research effectively to a general public. The thesis itself remains unchanged, as the grades were already determined at the A3. However, the quality of this lay presentation can still influence the final grade, potentially adjusting it up or down by half a point.

IV. Rubrics

IV.i. Rubric for A1

The rubric at A1 is based on the same criteria as the A3 rubric, see below. The supervisors judge whether a student is on track to obtain a *Green-light* at A3.

IV.ii. Rubric for A3

mark \ category	Research (50%)	Process (20%)	Communication (30%) (Report (80%) & Presentation (40%))
insufficient (<5,75)	- General problem cannot be explained - No specific research questions/objectives - Unable to place the research in a wider context, no clear literature research - The research resulted in almost no work using already existing sources - The results do not answer the research questions - No substantial conclusions	- Not autonomous or proactive at all - Never responsive when new alternatives are suggested - Rarely taking in feedback from supervisors and implementing changes - Misuse of resources (data, computational time, people time) - No real planning, missed most of the deadlines - No original ideas were provided within the project, most of the work is copied and already developed	- Report has no structure - Report does not document sufficiently the research done, not reproducible - Report lacks visual material - Presentation is chaotic, not clear structure - Presentation has no motivation - In presentation loses audience rapidly - Candidate cannot address the questions posed - Clear lack of understanding of the scientific problem under study
6	- Motivation can be broadly discerned, but it is not well understood - General problem is vague or without clear boundaries (scope) - Sufficient introduction and justification of the research topic, but superficial (limited literature review) - The choices of methods and data are not justified or explained - Limited critical attitude and ability to reflect on the wider scope of application of the research - The answers to the research questions are satisfactory - Results interpreted to a limited extent	- Sometimes autonomous and proactive, but generally needed steering by supervisors - Rarely came up with creative new ideas and new sources of information - Little response/action to feedback from supervisors for self-improvement - Makes inefficient but passable use of resources (e.g. tools, data, own/supervisor's time) - Contribution to the project is somewhat original - Limited initiative and suggestions within the project - Some initiative and suggestions by the student - Basic timeline and plan prepared, but little followed or updated	- Report has just right structure, consistency and clarity, with significant corrections by supervisors - Report does not document all the parts of the research done (reproducibility issues) - Presentation follows a structure, but with some issues in clarity - Presentation gives a decent summary of motivation, problem, work done, results and conclusions - Sufficient presentation material (e.g. slides, videos, demos) - Interaction with the audience is sufficient (eye contact, body language, tone of voice, pace of speaking) - Gets attention of the audience - Can answer most of the questions raised - Shows superficial knowledge, not in depth control of the topic
7	- Motivation can be understood and related to the problem - General problem is clear with defined boundaries (scope) - Sufficient introduction and justification of the research topic, with fair literature support (decent literature review) - The choices of methods and data are partly justified - Fair critical attitude and ability to reflect on the wider scope of application of the research - The answers for the research questions are more than satisfactory - Results interpreted with a critical attitude independently	- Mostly autonomous, generally trying approaches before asking for help - Few times came up with new ideas or found new sources of information - Was able to contribute to discussions about the research during meetings - Critical attitude towards the work done, but most key issues had to be pointed out by supervisors - Use feedback from supervisors for self-improvement - Use of resources is appropriate (e.g. tools, data, own/supervisor's time) - Contribution to the project is partly original - Some initiative and suggestions by the student - Good timeline and plan prepared, often followed or updated	- Report follows a structure, with issues in clarity and organization - Report documents all the parts of the research done (no reproducibility issues) - Report is generally well written, but contains significant errors and needs improvements - Abstract does not capture most of the work - Report properly acknowledges other work broadly and contains a fair list of references - Presentation follows a structure, but with some issues in clarity and organization - Presentation gives a decent summary of motivation, problem, work done, results and conclusions - Good presentation material (e.g. slides, videos, demos) - Interaction with the audience is appropriate (eye contact, body language, tone of voice, pace of speaking) - Gets attention of the audience and maintains it to some extent - Questions are answered well with some gaps - Confident with the content for its application
8	- Motivation is clearly shown and connected to the problem - General problem is clear and has defined limitations - Good introduction and justification of the research topic with supporting literature (but not all included) - The choices of methods and data are justified and logical - Demonstrate critical attitude and ability to reflect on the wider scope of application of the research - The answers to the research questions are good - Results interpreted critically and discussed in a broader scope of the discipline	- Mostly autonomous and proactive, generally taking control of the project and steering it to completion with some hiccups - Sometimes came up with new ideas and found new sources of information - Was able to contribute to lively discussions about the project during meetings - Critical attitude towards the work done, but key issues had to be pointed out by supervisors - Uses feedback from supervisors for self-improvement - Makes good use of resources (e.g. tools, data, own/supervisor's time) - Contribution to the project is original, with suggestions by supervisors - General initiative and suggestions within the project - Prepared a good and feasible plan at the beginning of the research project, which was mostly followed or adjusted when needed (e.g. according to progress and new findings)	- Report follows a structure, with minor issues in clarity - Report documents all the parts of the research done (no reproducibility issues) - Report is generally well written, but contains a few errors and needs improvements - Abstract captures most of the work - Report properly acknowledges other work most of the time and contains a mostly complete list of references - Work yields some other output (e.g. software, data), which is added to the report - Presentation follows a structure, but with some issues in clarity - Presentation gives a good summary of motivation, problem, work done, results and conclusions - More than satisfactory material (e.g. slides, videos, demos) - Interaction with the audience is good (eye contact, body language, tone of voice, pace of speaking) - Maintains attention of the audience for most of the presentation - Most questions are correctly answered - Very confident with the content at a research and development level
9	- Motivation is clearly described and connected with the need of solutions of the problem - General problem is clear, has boundaries or limitations and is feasible - Good introduction and justification of the research topic, with vast literature support - The choices of methods and data are justified and logical - Good critical attitude and ability to reflect on the wider scope of application of the research - The answers to the research questions are very good - Results interpreted critically and discussed in a broader scope of the discipline, with proposed solutions or alternative approaches when necessary	- Autonomous and proactive, taking control of the project and steering it - Most times came up with new ideas and found new sources of information - Was able to lead lively discussions about the research during meetings - Critical attitude towards the work done, pointing out the issues by him/herselves - Uses feedback from supervisors for self-improvement - Makes very good use of resources (e.g. tools, data, own/supervisor's time) - Contribution to the project is original, with almost no intervention by supervisors - Many initiative and suggestions within the project - Prepared a clear and feasible plan at the beginning of the research project, which was followed and improved when needed (e.g. according to progress and new findings)	- Report follows a clear structure - Report documents all the parts of the research done - Report is well written, with a very few writing errors - Abstract captures the essence of the work - Report properly acknowledges other work most of the time and contains a mostly complete list of references - Work yields some other output (e.g. software, data), which is added to the report and published in an ad hoc manner - Presentation follows a clear structure - Presentation gives a very good summary of motivation, problem, work done, results and conclusions - Very good presentation material (e.g. slides, videos, demos) - Interaction with the audience is very good (eye contact, body language, tone of voice, pace of speaking) - Maintains constant attention of the audience - Questions are answered well, without further deepening in the topic - Masters the content within the research topic
10	- Motivation is perfectly presented and connected with the need of solutions of the problem - General problem is clear, has boundaries or limitations and is feasible with the approach proposed - Excellent introduction and justification of the research topic, with all literature support - The choices of methods and data are justified, logical and the most efficient at the moment - Excellent critical attitude and ability to reflect on the wider scope of application of the research, making connection to simultaneous research performed by other peers - Results interpreted critically and discussed in a broader scope of the discipline, with proposed solutions or alternative approaches when necessary - The answers to the research questions are excellent - There is a clear evidence that the student is able to design new techniques or combine different techniques successfully in an innovative manner	- Highly autonomous and proactive throughout the process, taking full control of the project and steering it to completion in an efficient manner - Always came up with creative new ideas and found new sources of information - Was able to lead lively discussions about the research during meetings - Critical own attitude towards the work done - Actively uses both own discoveries and feedback from supervisors for self-improvement - Makes highly efficient use of resources (e.g. tools, data, own/supervisor's time) - Contribution to the project is original - Makes all initiative and suggestions within the project - Prepared an efficient, clear and feasible plan at the beginning of the research project, which was followed and improved when needed (e.g. according to progress and new findings)	- Report follows a clear and logical structure - Report thoroughly documents all the parts of the research done, which could be readily replicated using only the report as a base - Report is well written using clear scientific language and few errors - Report is visually appealing and uses figures and tables to best explain aspects of the research - Abstract captures the essence of the work - Report properly acknowledges other work everywhere and contains a complete and well-formatted list of references - Work attempts to yield other output (e.g. software, data) whenever possible, which is published following open science best practices (e.g. fully available source code on public repository with documentation and sample data) - Presentation follows a clear and logical structure - Presentation gives an easy to understand summary of motivation, problem, work done, results and conclusions - High-quality presentation material (e.g. slides, videos, demos) - Interaction with the audience is outstanding (eye contact, body language, tone of voice, pace of speaking) - Maintains constant attention of the audience - Questions are answered succinctly and with full awareness of the strengths and weaknesses of the research - Masters the content beyond the research topic

(Digital version in PDF and XLSX is available at <https://geomatics.bk.tudelft.nl/geo2021/rubric/>)

IV.iii. Rubric for A4

verbally	explained and argued for a lay audience without adjustments or in an incomprehensible way	explained and argued for a lay audience in an understandable way	explained and argued for a lay audience in an exceptionally good and engaging way
visually	shown and displayed for a lay audience without adjustments or in an incomprehensible way	shown and displayed for a lay audience in an understandable way	shown and displayed for a lay audience in an exceptionally good and engaging way
A4 final grade	A3 grade - 0.5	A3 grade	A3 grade + 0.5

V. Particular circumstances

V.i. Quorum at formal assessments

A quorum is required for each assessment to be valid.

- For A1: responsible supervisor, second supervisor and delegate of the Board of Examiners.
- For A3: responsible supervisor, second supervisor, co-reader and delegate of the Board of Examiners.
- For A4: responsible supervisor and second supervisor.

In case of absence, action will be taken as follows:

Absence of the delegate of the Board of Examiners The Board of Examiners appoints the delegate of the Board of Examiners and the substitute delegate of the Board of Examiners for all students. If the delegate of the Board of Examiners cannot attend an assessment, the substitute will be asked and the Secretary of the Board of Examiners will be informed on this replacement. The substitute delegate of the Board of Examiners is registered in My Case.

Absence of one of the supervisors and/or the co-reader If it is known in advance that a supervisor or the co-reader will be unable to attend, a presentation must be held for that assessor prior to the evaluation. The assessment of that assessor must be written down with comments and feedback. This assessment must be given to the delegate of the Board of Examiners in signed letter within a closed envelope or sent by e-mail. At the deliberation, this assessment will be taken into account by the other relevant members of the supervisory team for determining the final assessment. In case of an unexpected absence, the other supervisory team members will search for a replacement. The Secretary of the Board of Examiners is also informed by the responsible supervisor or the delegate of the Board of Examiners about this absence. The assessment should preferably be continued, but the final assessment should be determined after hearing the absent supervisor. The determination for a 'go' or 'no go' at A1 or A3, including the registration of the grades on the list at A3 can only take place after consulting the absent assessor. If this is not possible, the final assessment is postponed. In both cases, a meeting with the absent supervisor takes place on the shortest possible term, to determine a final conclusion. In case of doubt or on request of the student, it may be decided that an extra assessment must be held.

V.ii. Difficulties at the deliberation

It may occur that the deliberation does not lead to an unambiguous assessment. The delegate of the Board of Examiners informs the student on this situation and explains the applied procedure and the corresponding terms. Subsequently, the presented products will be handed over to the Secretary of the Board of Examiners. The Secretary of the Board of Examiners decides which member of the committee will look after this matter. This member of the Board of Examiners will reconvene the supervisory team and the delegate of the Board of Examiners for a redeliberation, chaired by the member of the Board of Examiners, in which

an attempt is made to reach consensus. If unexpectedly this does not succeed, the member of the Board of Examiners will make a final decision.

V.iii. Plagiarism

The plagiarism scan on graduation documents is implemented for all graduating students, to guarantee the authenticity of the graduation work. For this, the Turnitin tool in Brightspace is used. The tool will make it easier for the student and supervisors to check the work of a student on originality and plagiarism. The supervisors and delegates will be enrolled by Education and Student Affairs in the Plagiarism Brightspace course. The student has the possibility to upload provisional versions of their document as often as desired for plagiarism feedback, only meant for the student. The final version of the Graduation Report for the A3 will be submitted in the final version folder of the plagiarism scan, at the latest one week before the A3. The final submission folder will only allow one submission for each student and the plagiarism feedback will only be visible for the supervisors. It is the responsibility of the responsible supervisor to determine whether the results of the plagiarism scan in the final folder are an indication of actual plagiarism. In all cases, suspicion of plagiarism or not, the responsible supervisor should share the findings with the student, the second supervisor, the co-reader and the delegate at the A3. If there is a suspicion of intentional plagiarism, the responsible supervisor should discuss this with the student and notify the Board of Examiners afterwards.

VI. Data management checklist

This checklist is relevant for all graduation projects at the Faculty of Architecture and the Built Environment. The form is intended to highlight common aspects of graduation projects that require particular attention with regard to planning the research and data management. Relevant information and supplementary sources regarding each question are provided below each question.

With this checklist, the faculty wants to avoid that students unexpectedly find themselves in complex and stressful situations, in which ethical or privacy matters and/or other laws and regulations become an issue. In projects involving humans, certain types of data processing increase the risks to the human participants: planning such projects requires additional evaluations and advice from university staff before ethical approval can be received and the project can begin. In the case of a graduation project, obtaining additional advice or permits may delay the project with an extra education period or semester. To avoid this, it is recommended that students set up a graduation project with a low level of risk. Therefore, all students have to check their risk, by completing this checklist before their A1.

The first section of the checklist (A) should be completed by all students, together with their supervisor, during the planning of the graduation project, before the A1. It does not need to be submitted to anyone for review or approval. Please consider questions 1 to 3 carefully in relation to the intended graduation project, and answer with ‘yes’ or ‘no’.

The second section of the checklist (B) should only be completed if the graduation project involves working with data from human participants. In that case, the student and their supervisor must apply for and receive ethical approval from the [Human Research Ethics Committee](#) (HREC) before the project can begin (see the paragraph ‘Explanation and follow-up’ after the questions). The student can submit the application to the HREC, but the supervisor is responsible for making sure that the project is compliant with relevant privacy regulations and ethical policies.

	Section A. General considerations	yes	no
1	Is the graduation project conducted as part of an internship (at a company), or as part of a research project at TU Delft? If a student’s graduation project is conducted at a company or as part of a research project at the university, questions of data ownership and intellectual property rights need to be addressed in a written graduation or internship agreement before the project begins. Students and their supervisor should consult the Intellectual Property Rights of Students webpage. Additional information can also be found in the Extended Personal Research Data Workflow		
2	Does the project involve conducting (part of) the research outside the Netherlands? Students who intend to travel abroad (even to other EU countries) for		

	Section A. General considerations	yes	no
	study, exchange, research, internship, or graduation project purposes need to follow the Travel Safety Protocol . This includes attending a mandatory Travel Safety Training Session: see the Disclaimer		
3	Will the research involve processing data from humans, such as running a survey, conducting interviews or workshops, collecting data through social media or internet forums, or re-using existing datasets about humans provided by a third party? (If 'yes', see follow-up questions 4 to 13 in Checklist B.) Students who work with data from human participants must complete the next section and apply for and receive ethical approval from the Human Research Ethics Committee (HREC) before conducting the research.		

	Section B. Extended risk factors (only if question 3 has been answered with 'yes'.)	yes	no
4	Will the project involve participants who may be considered vulnerable, such as the elderly, refugees or asylum seekers, ethnic minorities, patients, or people with disabilities? Participants who may suffer very adverse consequences (for instance, due to discrimination) if their personal data became publicly available can be considered vulnerable.		
5	Will the project involve participants who cannot themselves give informed consent for taking part in the project, but for whom consent must be obtained from a legal guardian? Participants who cannot give informed consent can include, for instance, children or participants with intellectual disabilities, mental disorders, or dementia. Such participants are also considered vulnerable in the context of the General Data Protection Regulation (GDPR).		
6	Will the project involve processing any of the special categories of personal data below? • Race • Ethnicity • Criminal offence data • Political opinion • Union membership • Religious or philosophical beliefs • Sex life and/or sexual orientation • Health data (including measurements such as heart rate)		

	Section B. Extended risk factors (only if question 3 has been answered with 'yes'.)	yes	no
	• Biometric or genetic data (including fingerprints, iris scanning, facial recognition) The General Data Protection Regulation (GDPR) defines stricter rules for processing special categories of personal data. If it is necessary to process these data in a project, it is important to provide additional safeguards.		
7	Will the project involve processing personal data that could be considered sensitive, such as the ones listed below? • Information about a person's income, debts, or other payments • Information about a person's (un-)employment status • Information about a person's performance at school or work • Information about relationship problems or (gambling) addiction • Information about poverty, domestic violence, or youth welfare/ social work involvement Some types of personal data are considered sensitive, because they can have a high impact on the privacy of the data subject if other persons gain access to these data. Sensitive personal data should only be processed if necessary: in such cases, additional safeguards need to be put in place.		
8	Will the project involve processing video-recordings, or photographs of participants? TU Delft considers photographic and video-materials of research participants to be sensitive personal data. If such data need to be processed, additional safeguards must be put in place.		
9	Will the project involve sharing or transferring personal data between multiple partners or collaborating organisations involved, such as between TU Delft and an internship company? According to privacy law, sharing personal data between organisations requires a privacy agreement to be in place: setting this up takes time, and requires support from additional university staff. Furthermore, personal data sharing can potentially expose research participants to different types of risks: these risks must be considered in the ethical application.		
10	Will the project involve deception, or covert observation of participants?		

	Section B. Extended risk factors (only if question 3 has been answered with ‘yes’.)	yes	no
	In some types of research, obtaining informed consent for processing participants’ personal data is not an option: for instance, if the research involves deception, or the research is covert (conducted without participants knowing about it). In such situations, the steps to mitigate risks to participants are important, and an alternative legal basis for processing the participant’s data needs to be established with the help of additional support staff.		
11	Will the project involve working with social media data? Social media data are personal data, but since it is usually not possible to ask for informed consent for processing social media data, another legal basis for processing the participant’s data needs to be established. Processing of social media data also involves legal considerations related to terms of use of data from third-party platforms: therefore, research with social media data requires expert support on privacy, ethics, and legal matters.		
12	Will the project involve using learning algorithms or other AI to analyse, combine, or otherwise process data from participants? The use of AI in research involves many considerations in terms of data protection, ethics, security, and intellectual property: for more information, see TU Delft’s Instructions for use of Generative AI .		
13	Will the project involve participants who are based in a country or countries outside of the EU? Students affiliated with TU Delft must comply with Dutch and EU regulations of personal data processing (GDPR). Furthermore, the student and their supervisor must make sure that the research complies with local (privacy) legislations of any foreign destinations. Additional support from an external (local) expert may be required.		

Explanation and follow-up

If you have answered ‘no’ to all questions 4 to 13, your project is likely to be considered low or minimal-risk: see the paragraph ‘Projects with minimal or low-risk’ on the next page.

If you have answered ‘yes’ to one or more of the questions 4 to 13, your research likely involves extended or high risks to participants, according to the [General Data Protection Regulation](#) (GDPR) and TU Delft’s privacy and ethical policies: for information regarding such projects, see the paragraph ‘Projects with extended or high-risk’ on the next pages.

Projects with minimal or low-risk

If you have answered ‘no’ to questions 4 to 13, your project is likely to be considered low-risk. This does not mean that the project involves no risks at all, but suggests that these risks can likely be addressed by the student and supervisor in the application to the [Human Research Ethics Committee](#) (HREC) within the timeline for a graduation project and without need for additional support.

Compiling the HREC application:

An application to the HREC generally involves a Data Management Plan (DMP), a risk-identification and mitigation checklist, and informed consent materials. Master’s students at ABE who intend to compile a HREC application are advised to make use of the following support documents:

- the [student guide](#)
- the [Example Data Management Plan](#) for MSc projects

The graduation supervisor is [responsible](#) for the student’s project and ethical application, and must provide support for compiling the HREC application documents.

Additional support

For low-risk student graduation projects, compiling of the HREC application documents should be done by the student in consultation with the supervisor. The Faculty Data Steward can be contacted for individual questions [at datasteward-BK@tudelft.nl](mailto:datasteward-BK@tudelft.nl) however, the Data Steward does not provide detailed feedback on student DMPs for low-risk HREC applications.

Additional resources

The HREC has guides available for [completing the checklist](#) and for compiling [informed consent](#) materials.

Additionally, the [Guide to the Extended Personal Research Data Workflow](#) has been created to help researchers and students who work with human participants comply with both GDPR principles and TU Delft’s policies on Data Management and Human Research Ethics.

Timeline

Minimal or low-risk HREC applications are generally processed faster than extended or high-risk applications (see the paragraph below). Nevertheless, the initial evaluation by the HREC usually takes approximately 2 weeks, and may take longer during busy periods or holiday: see the [HREC website](#) for up-to-date information. Additionally, the application may require revisions before final approval is granted. If you do not receive an initial response about your ethical application after 4 weeks from the time of submission, you may follow up with the HREC to enquire about an update.

Projects with extended or high-risk

If you have answered ‘yes’ to one or more of questions 4 to 13, there are potential increased risks related to how data from human participants will be processed in your project. These risks will need to be addressed in consultation with the Data Steward and other relevant support staff before submitting the ethical application to the [Human Research Ethics Committee](#) (HREC)

Compiling the HREC application

An application to the HREC generally involves a Data Management Plan (DMP), a risk-identification and mitigation checklist, and informed consent materials. Master's students at ABE who intend to compile a HREC application are advised to make use of the following support documents:

- the [student guide](#)
- the [Example Data Management Plan](#) for MSc projects

The graduation supervisor is [responsible](#) for the student's project and ethical application, and must provide support for compiling the HREC application documents.

Additional support

Once the DMP has been compiled and reviewed by the supervisor, feedback should be requested from the Data Steward via DMPonline. After this, any other necessary support staff will need to be contacted. Crucially, if the project involves one or multiple ways of personal data processing that could result in high-risk to the participants according to the GDPR, the TU Delft Privacy Team must

be consulted to establish whether or not a [Data Protection Impact Assessment](#) (DPIA) is required.

Additional resources

The HREC has guides available for [completing the checklist](#) and for compiling [informed consent materials](#). Additionally, the [Guide to the Extended Personal Research Data Workflow](#) has been created to help researchers and students who work with human participants comply with both GDPR principles and TU Delft's policies on Data Management and Human Research Ethics.

Timeline

It can take a long time to compile a complete research plan and HREC application for projects involving extended risks. DMP feedback from the Data Steward usually takes around 2 weeks, but can take longer during busy periods or holidays. Receiving additional support from other staff, such as the Privacy Team, can take anywhere from a few days to multiple weeks, depending on the project and capacity of university staff. If a DPIA is deemed necessary, it can take anywhere from 4 weeks to several months.

It is important to note that advice from the Privacy Team or other support staff, as well as any additional documents (such as necessary contracts, or a DPIA, if needed) must be in place before the application is submitted to the HREC. The initial evaluation by the HREC can be processed in 2 weeks, but may take longer during busy periods or holidays: see the [HREC website](#) for up-to-date information. Additionally, the application may require revisions before final approval is granted. If you do not receive an initial response about your ethical application after 4 weeks from the time of submission, you may follow up with the HREC to enquire about an update.

Considering the limited time available for students conducting their graduation projects, students working with data from human participants are strongly advised to prioritise low-risk research projects. If a student project necessitates processing data in ways that are considered extended or high-risk, both student and supervisor need to be aware of the extended processing times involved in obtaining ethical approval and beginning the graduation project.

VII. Final attainment levels

VII.i. Generic Final Attainment Levels (from TU Delft)

A student graduating with a Master's in Geomatics:

G1	Is skilled in one or more academic disciplines. A university graduate is familiar with existing academic knowledge and is capable of expanding this through study.
G2	Is able to perform research. A university graduate is capable of acquiring new academic knowledge through research. In this connection, 'research' means developing and discovering new knowledge and insights in a purposeful and methodical manner.
G3	Is able to design. Many university graduates will formulate designs in addition to performing research. Designing is a synthetic activity geared towards bringing about new or modified artefacts or systems, for the purpose of creating values in accordance with the requirements and wishes stated beforehand (for example, mobility or health).
G4	Has an academic approach. A university graduate follows a systematic approach, characterised by developing and using theories, models and coherent interpretations, has a critical attitude and has an understanding of the unique nature of science and technology.
G5	Possesses basic intellectual skills. A university graduate is able to reason, reflect and form judgments. These are skills which are learned or sharpened in the courses of a discipline, and are generically applicable afterwards.
G6	Is able to cooperate and communicate. A university graduate has the capacity to work with and for others. This requires not only sufficient interaction, a sense of responsibility, and leadership, but also good communication with individuals both in and outside the profession. In addition, the university graduate is able to participate in academic or public debates.
G7	Takes into account the temporal and social context. Science and technology are not isolated, but rather, always have a temporal and social context. Opinions and methods have a background; decisions have social consequences over time. A university graduate realises this and is able to integrate these insights into their academic work.

VII.ii. Specific final attainment levels for the Master's Geomatics

A graduate of the Master's programme in Geomatics:

GM1	Is familiar with existing scientific knowledge, and has the competence to increase and develop this through study, and is competent in one or more scientific disciplines covering the complete geo-information chain, by:
------------	--

	a) having appropriate knowledge of the role of geo-information in decision making applied to the built environment b) applying a formal, theoretical and methodologically sound approach to store, process, manage, disseminate and visualise geo-data and geo-information c) acting at the forefront of data analysis methodology and spatio-temporal database management systems technology d) having appropriate knowledge of legal and organisational aspects of geo-information; and e) utilising a technology and application driven approach to geo-data acquisition and geo- information extraction.
GM2	Has the research competences to: a) investigate and analyse spatio-temporal models and systems of complex real-world situations b) formulate hypotheses or research questions and develop new concepts c) validate developed concepts by appropriate prototypes, and d) analyse and evaluate thoroughly obtained results.
GM3	Has the competences to design, implement and use: a) models and systems for management, storage, processing, integration, exchange and dissemination of geo-information b) systems and methodologies to analyse measurements and other evidence material for the purpose of geo-information extraction; and c) select, combine and extend above-mentioned systems and methodologies to meet geo information requirements, notably in the built environment, in the present and in the future.

